

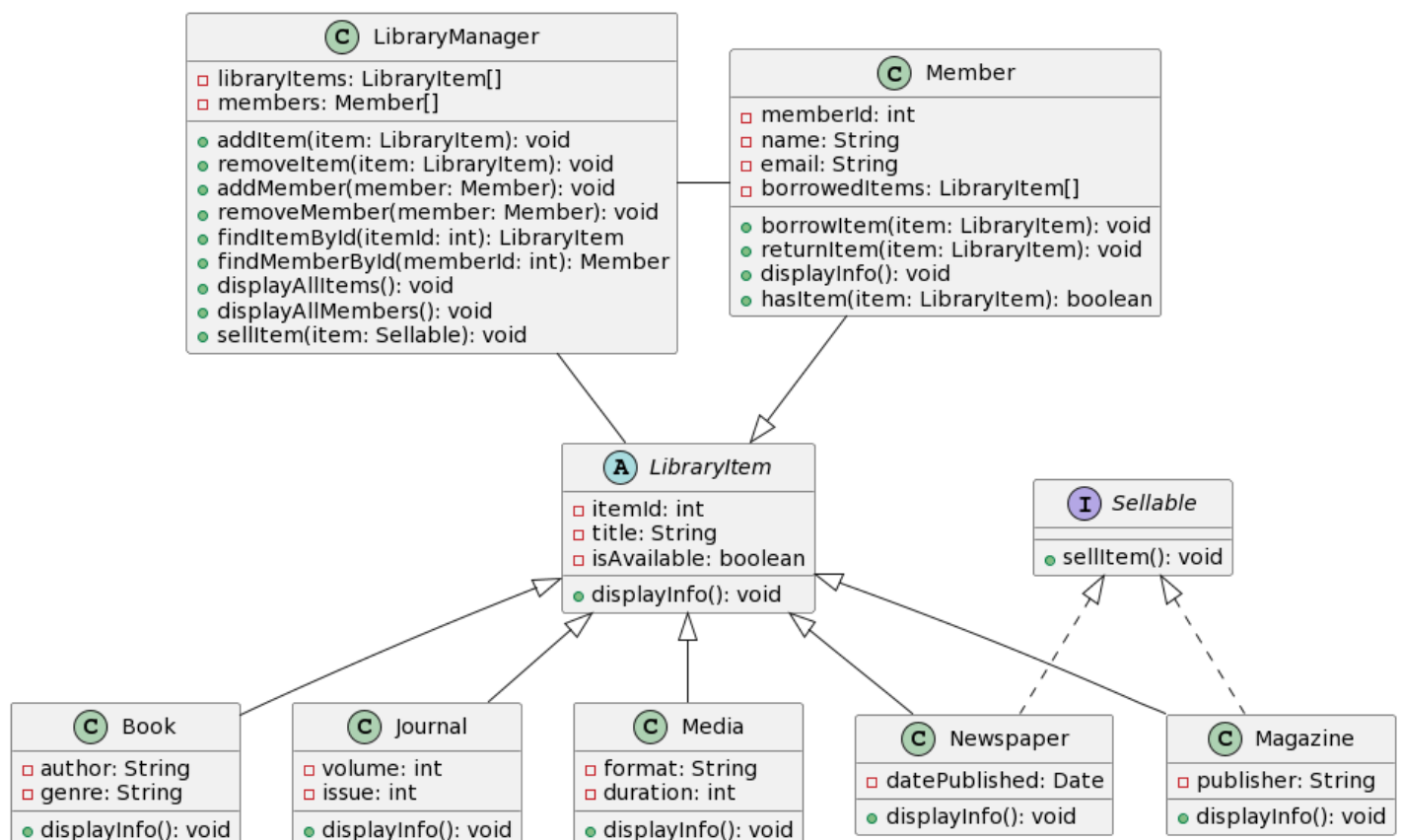
CSE 231s Advanced Computer Programming Project

Select one of the following applications ideas then design and implement an application that follows the Object-Oriented Programming Paradigm:

- Online-Language-teaching system: with different types of **languages**, users, quizzes, ...
- Hotel-Booking system: with different types of **rooms**, users, bookings, offers,
- Pharmacy Management System: Managing pharmacy **items**, orders, recipes, and payments.

Your design should follow the class design guidelines (As explained in chapter 13) . The design should show your ability to use inheritance and polymorphism using concrete or abstract super classes. In addition, you should have at least one interface. The **bold** objects (languages/ rooms/ items) in your implementation should be sorted using the generic sort method.

Here is a possible design for a library management system for you to understand how to start (do not use this design or this application):



- Number of students per group: up to 5 students
- Grading Criteria:
 - The presentation of your design
 - The application of all required OOP concepts required in the project description.
 - Explaining your role in the project and how you applied the OOP concepts in the project.
 - Being able to answer questions about extensions that can be added to the projects with minimal changes in the code.
 - The correctness of the project functionality
- Grading Scheme
 - Grade 3: Implementation and proper discussion of the applied OOP concepts and design guidelines
 - Grade 4: Grade 3 requirements + Implementation and proper discussion of the exception handling
 - Grade 5: Grade 4 requirements + Implementation and proper discussion of the UI (User interface)
- More details about the presentation and demo requirements will come later.
- Deadline: Last week before exams (More details will come later)